AI101 Mini-Project | Team 7

Calories Burnt Prediction

Predicting how many calories you burn in a workout, using duration, heart rate, body temperature and age

Team and roles

Member	Entry no.	Role
Piyush Tiwari	2026ESB1487	Documentation lead
Devansh Singh Vardhan	2026EEB1342	Execution lead
Avinash	2026AIB1025	PPT lead
Sunnam Jagadeesh	2026MCB1575	Video lead

Problem and data

What and why

- Goal: predict calories burned from duration, heart rate, body temperature and age
- Why: gym machines and smartwatches only give rough estimates, so we let a model learn from real workout records
- Type: regression, since the answer is a number

Dataset

Kaggle: Calories Burnt Prediction (Ruchika Kumbhar)

500 rows

9 columns

Columns: User_ID, Gender, Age, Height, Weight, Duration, Heart_Rate, Body_Temp, Calories (target)

Approach

1 Data

Load the CSV from the data folder

2 Clean

Check for missing values and look at the data with some plots

3 Model

Train on 80% of the data and test on the other 20%

4 Output

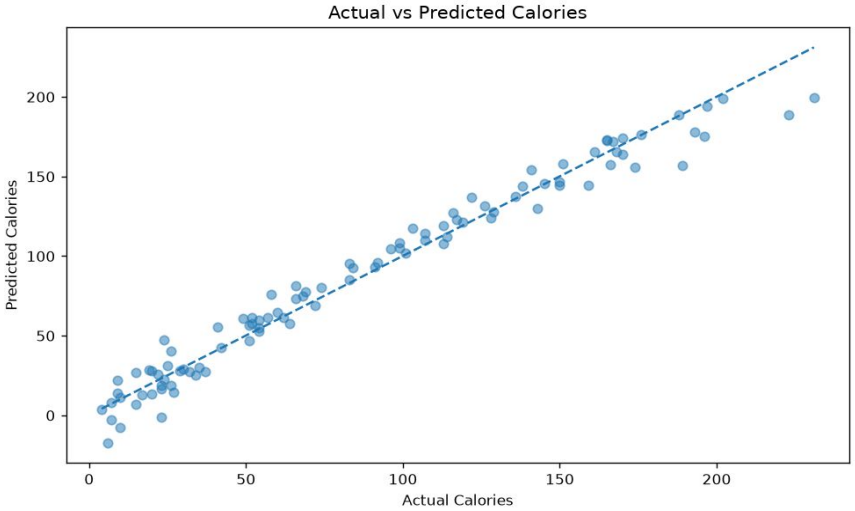
Get the predicted calories, accuracy and plots

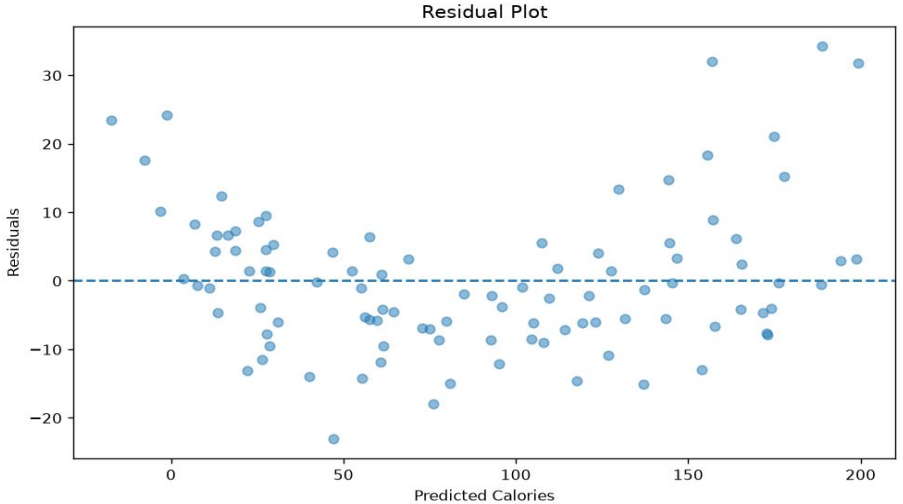
Libraries used

pandas **numpy** **matplotlib** **scikit-learn**

Model we used : Linear Regression

Results





0.969

R^2 score

8.45

MAE

10.72

RMSE

```
========== MODEL PERFORMANCE ==========
Mean Absolute Error (MAE): 8.45
Mean Squared Error (MSE): 114.96
Root Mean Squared Error (RMSE): 10.72
R2 Score: 0.9693
R2 Score (%): 96.93 %
```

Conclusion & Project Details

What worked

- Duration and heart rate gave the strongest predictions
- Splitting into train and test showed us the real accuracy
- Linear Regression scored $R^2 = 0.969$

Who did what

Documentation: Piyush

Execution: Devansh

PPT: Avinash

Video: Jagadeesh

All four wrote and committed code

Repository: https://github.com/piyush-esb1487/Athlete-calorie-counter